

L'indagine di contesto *principi*

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Gathering Customer Data

- Why?
 - Many Problems with Software / IT Systems Can Be Traced Back to Problems With Requirements
 - Also True of Other Types of Systems
- Theory
 - Better Customer Contact
 - Better Data
 - Better Design

What Sort of Data Do We Want?

- Accurate Data
- Appropriate Data
 - Relevant to Design
 - Not Just Averages / Aggregate Data
 - e.g. Marketing Data Not Always Useful:
May Say What People Will BUY,
Not How To Design It
- Detailed Data
 - Specific Enough to Answer Design Trade-Offs
 - e.g. Marketing Data Usually Pretty Broad

Why Is Gathering Customer Data Hard?

- Customers Focused on Current Operations
 - Tend to Suggest Small Fixes
 - May Not Have Ever Thought of Significantly Different Way of Doing Task
- Customers Tend to Give Partial Input
 - Tendency to Focus on Parts of Task They Have to Think About
 - Aren't Aware of All the Contextual Factors
 - Aren't Aware of All Their Skills
 - Procedural vs Declarative Knowledge

Example of Unnoticed Data

“An entire project team hangs out in the hallway outside their offices every morning and chats over coffee and donuts. Does anyone know that this is a critical project coordination meeting?”

- If you don't catch this essential fact, your information system to enable telecommuting may be unsuccessful.

Why Is Gathering Customer Data Hard? II

- Customers Don't Get Very Detailed
 - Tend to Generalize Things
 - Often Can't Even Explain Them Out of Context
- Customers May Not Be Very Accurate
 - Qualitative Statements
 - Relative Statements (e.g. Too Much Work!)
 - Inaccurate Estimates of Quantities

Example of Poor Data

- Try Getting Someone to Explain To You In a Lecture Hall How They:
 - Fly a Plane
 - Drive a Car
 - Run a Warehouse
- Domain Experts Are Not Necessarily Good Communicators.
- Communicating Out of Context Is Difficult.

Why Is Gathering Customer Data Hard? III

- Surveying All The Customers.
- Who Is Your Customer?
- Customers Often Have a 'Frame of Reference'
 - Engineers Will Suggest Technical Fixes...
 - Managers Will Suggest Regulatory Fixes...
 - Workers Will Suggest Procedural Changes...
- You Have a 'Frame of Reference'.
- You Already Are Biased by Design Ideas.

A Cautionary Note...

- Don't Get Ahead of Yourself
- Your Design Ideas Should Be Driven By Data --
Not Vice-Versa!
- But It's So Tempting To Sketch Out Some Design Ideas
And Show It To Them...

Purpose of Contextual Inquiry

- Get Data Suitable For Design
 - Understand Structure of Work Practice...
- Get Concrete Understanding of Domain Experts' Unarticulated Knowledge and Skill
- Get Suitably Detailed Data
 - Don't Accept Generalizations and Abstractions

'Textbook' Contextual Inquiry

- Each Interview 2-3 Hours
- 10-20 Interviews (or more)
- Follow-Up Design Meetings
- Practices
 - Context: Go To Customer's Workplace
 - Partnership: Talk With Them and Engage Them
 - Interpretation: Develop a Shared Understanding
 - Focus: Direct the Inquiry To Meet Design Needs

Can't Always Get This!

- Resource Issues
- Intrusiveness / Safety
- Lack of Partnership
- So Compensate!
 - Try to Situate Interviewees In Context
 - Use Videotapes and Artifacts
 - Become (Partial) Domain Experts

Conducting an Interview

- No One Hard-And-Fast Rule
- No Matter How Curmudgeonly the Interviewee, Remember That You Are the Interviewer
 - Think About an Appropriate Relationship Model...
 - Scientist/Subject **No**
 - Interviewer/Interviewee **No**
 - Expert/Novice **No**
 - Guest/Host **No**
 - Master/Apprentice! **Yes!**
 - They Demonstrate, You Investigate, They Explain...
 - There Will Be Times You Need to Deviate A Bit...
 - And Play The Part!
 - Won't Always Work, For Reasons Outside Your Control

Note What They Say and What They Do

- Note What They Say
 - Declarative Knowledge
- Note What They Do
 - Ask About It
 - Create Opportunities To Examine Procedural Knowledge
 - Create Opportunities To Examine Habitual Behavior
 - Create Opportunities To Get Greater Detail

An Interesting Observation I

“One customer said he would not use a manual’s index to find the solution to a problem: “It’s never in the index.” He could not say what led him to this conclusion, what he had looked up and failed to find. All his bad experiences were rolled up into one simple abstraction: it’s not there. But when we watched him looking things up, we could see that he was using terms from his work domain, but the index listed parts of the system.”



An Interesting Observation II

“A customer was unable to describe how she made her monthly report. When asked to create it, she pulled out her last report and started filling in the parts.”

Talking About Work While Doing It

- Allows You To Quickly Get Past Generalizations
- Allows You To See The Entire Context
- Allows You To Develop Some Domain Expertise (and Intuition for Design)
- Also Makes Customers Happier



Principles of Contextual Inquiry

- Context
- Partnership
- Interpretation
- Focus

First Principle of CI

- Context: “Go to the Customer’s Workplace and See the Work As It Unfolds”
 - Summary vs Ongoing Experience
 - Get Around Generalizations!
 - See Things From Multiple Frames!
 - Concrete Data vs Abstract Data
 - Normally Abstraction Is a Valuable Skill...
But Does the Customer Have the Right Abstractions?
 - Get Concrete Data
 - Get Artifacts
 - Don’t Just Aggregate Things

Second Principle of CI

- Partnership: “Make you and the customer collaborators in understanding her work.”
 - Use your relationship model to tilt the roles to one that gets them to give you information
 - Control the interview *process* -- don't control what data is given
 - Play a bit with the relationship model -- you need data (good apprentice) and also insights into practices (more observant partner, educator)
 - Withdrawal and return...
 - Describe technology...

Third Principle of CI

- Interpretation: “giving meaning to observations”
 - Change ‘data’ into ‘information’
 - Examine it from many different frames
 - Look for structure, work patterns, etc.
 - From facts, make hypotheses about implications, test them out -- and look at implications for design
 - But don’t bias them -- and listen carefully!

“It’s like a travelling office”, you say, looking at how a salesman has set up his car. “Well -- like a traveling desk,” he responds.

Fourth Principle of CI

- Focus: “keep the conversation on useful topics without wresting control entirely away from your customer/expert”
 - Remain objective, on track
 - Like steering a conversation -- gently!
 - Think of it as:
 - Getting more detail
 - Covering all relevant aspects of the domain
 - Focus should be pre-specified
 - Capitalize on surprises! Admit ignorance!

Contextual Interview Structure

- Conventional Interview
 - Administrative Stuff
- Transition
 - Explain the Ground Rules
- Contextual Interview Proper
- Wrap-Up
 - Back-up, Back-out -- What Didn't You Cover?

Contextual Inquiry - approfondiamo...

So What is Contextual Inquiry?

- A ***Process*** by Which To Gather Customer Data with the Following Objectives:
 - Get Data About the Structure of Work Practice
 - Make Unarticulated Knowledge Explicit
 - Capture Details of Habitual, Invisible Work

Before Interview: Decide on a Focus

- Focus: 4th Principle of Contextual Inquiry
- Should Be Decided Upon Before Interview(s)
- Will Change As Project Evolves
 - Narrower or Broader...
 - Different Reference Frames
 - Different Subject Matter
- Look for Metaphors
 - And Use Them as Data Collection Tools

Metaphors

- E.g. You're Tasked With Making an On-Line Search and Retrieval System
 - Examine How People Do Search and Retrieval in a Number of Domains
 - Physical (e.g. Library)
 - On-Line
 - Use These Outside Studies to Bring Insights to Topic at Hand

Variations in Focus I

- Designing a Known Product
i.e. Something Already Known and Accepted in the Workplace
 - Expectations Already Exist
 - Competitive Products for Comparison
 - Customer Feedback on Good-and-Bad
 - Need to Add More:
 - More Functionality
 - More Usability

Variations in Focus II

- Addressing a New Work Domain
i.e. A Work Domain Created By New Technology and Practices (e.g. Telecommuting, Word Processing, etc.)
 - More Difficult -- No Current Practice to Study
 - Can Study Current Domains, Though!
 - Much of Results of Your Design Will Be Driven By:
 - Intent and Goals
 - Inherent Structures to Tasks
 - Information Flow
 - Solutions to Historic Technological Limitations

Variations in Focus III

- New Technology
i.e. How Can I Apply Technology to New Domain?
 - Look at Fundamental Needs of the Domain
 - Do They Need Technology?
 - How Could the Technology Be Used to Improve Things?
 - Look at Metaphors
 - Look at Analogous Situations

Variations in Focus IV

- Designing Information Technology
i.e. Bring Technology Into Data Flow and Coordination Activities of an Organization
 - Implementing IT Requires Knowledge:
 - How Activities Can Be Integrated
 - ‘Within’ a Single Job
 - Within an Organization
 - How Information Can / Should Flow
 - What Technology Already Exists
 - What Work Processes Would Work Well

Before Interview: Who to Interview?

- Who Is Your Customer?
- Who Is The Worker?
- How Much Variation Is There?
 - Culturally
 - Allowably (e.g. Do Workers Have Much Freedom?)
 - Think Between and Within Departments
- How Much Variation Should You Support?

During Interview: Interview Structure

- Conventional Interview
 - Administrative Stuff
- Transition
 - Explain the Ground Rules
- Contextual Interview Proper
- Wrap-Up
 - Back-up, Back-out -- What Didn't You Cover?

Can We Always Follow This Structure?

- Uninterruptible Task - NO
 - Intermittent Task - NO
 - Extremely Long Task - NO
 - Internalized Task - NO
 - Extremely Focused Task - NO
 - 'Normal' Task - YES
- Fortunately, We Have Other Methods
 - Video tap and post-hoc protocol
 - Longitudinal Study
 - Artifact-based verbal protocol
 - Explanation of the process
 - Analysis of data
 - Modeling of Processes



Factors Impacting Interpretation

- Frames of Reference
- Biases
i.e. HOW You Look At Customer Data
(and How Customer Looks At Themselves)

Also Keep Track of the Goals



"You must have got it wet!"

Fonte

Jennifer Ockerman, Georgia Institute of Technology
Atlanta, GA USA

Testo:

Hugh Beyer and Karen Holtzblatt, 1998, *Contextual Design:
Defining Customer Centered Systems*, Morgan Kaufmann
Publishers

Cognitive maps

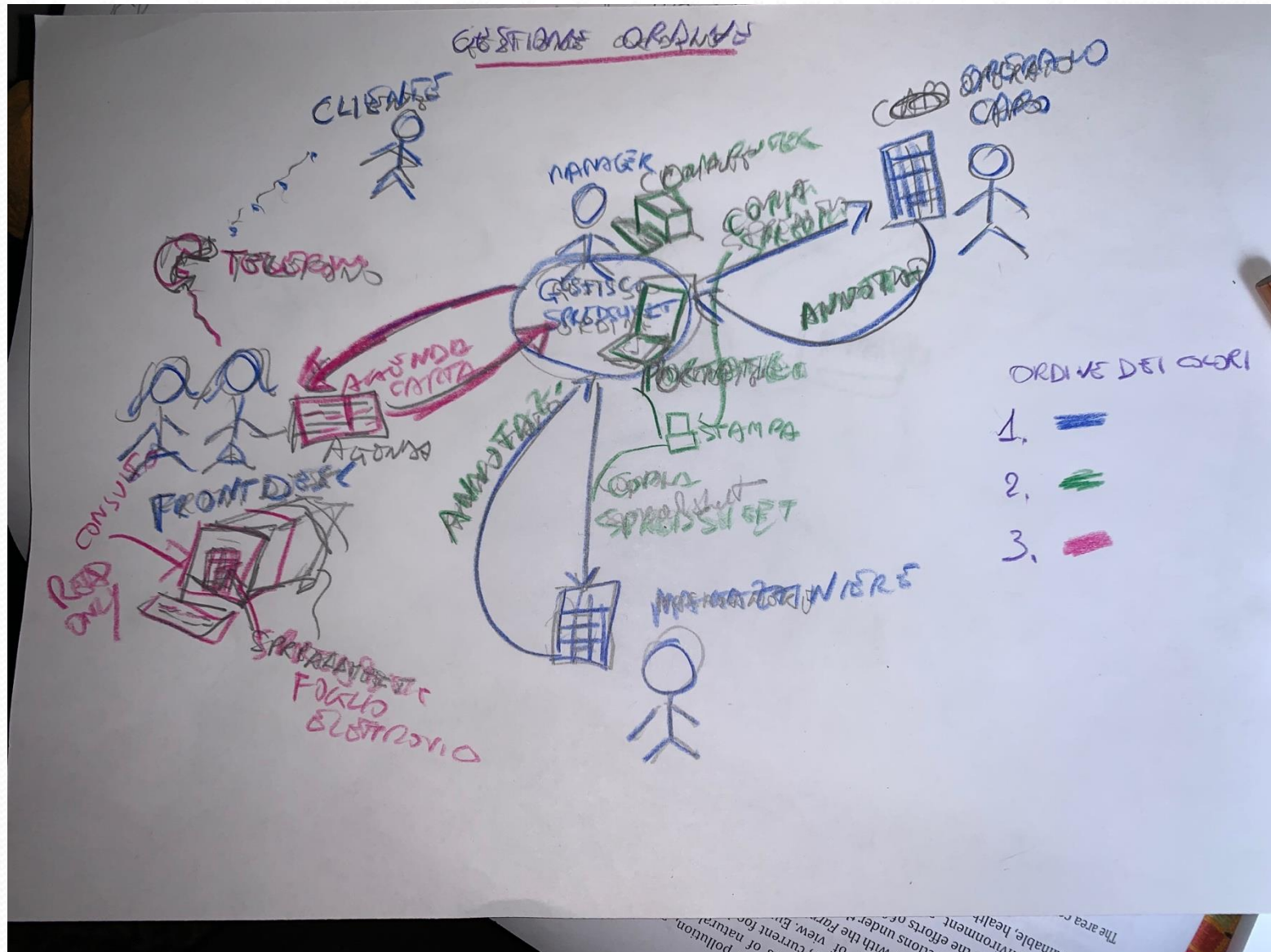
- starts with a question about an activity
 - ❖ ask about “securing your house” or “learning to speak French.”
- Limited to geographic locations but can include websites, devices, and books

Esercizio sulle cognitive maps

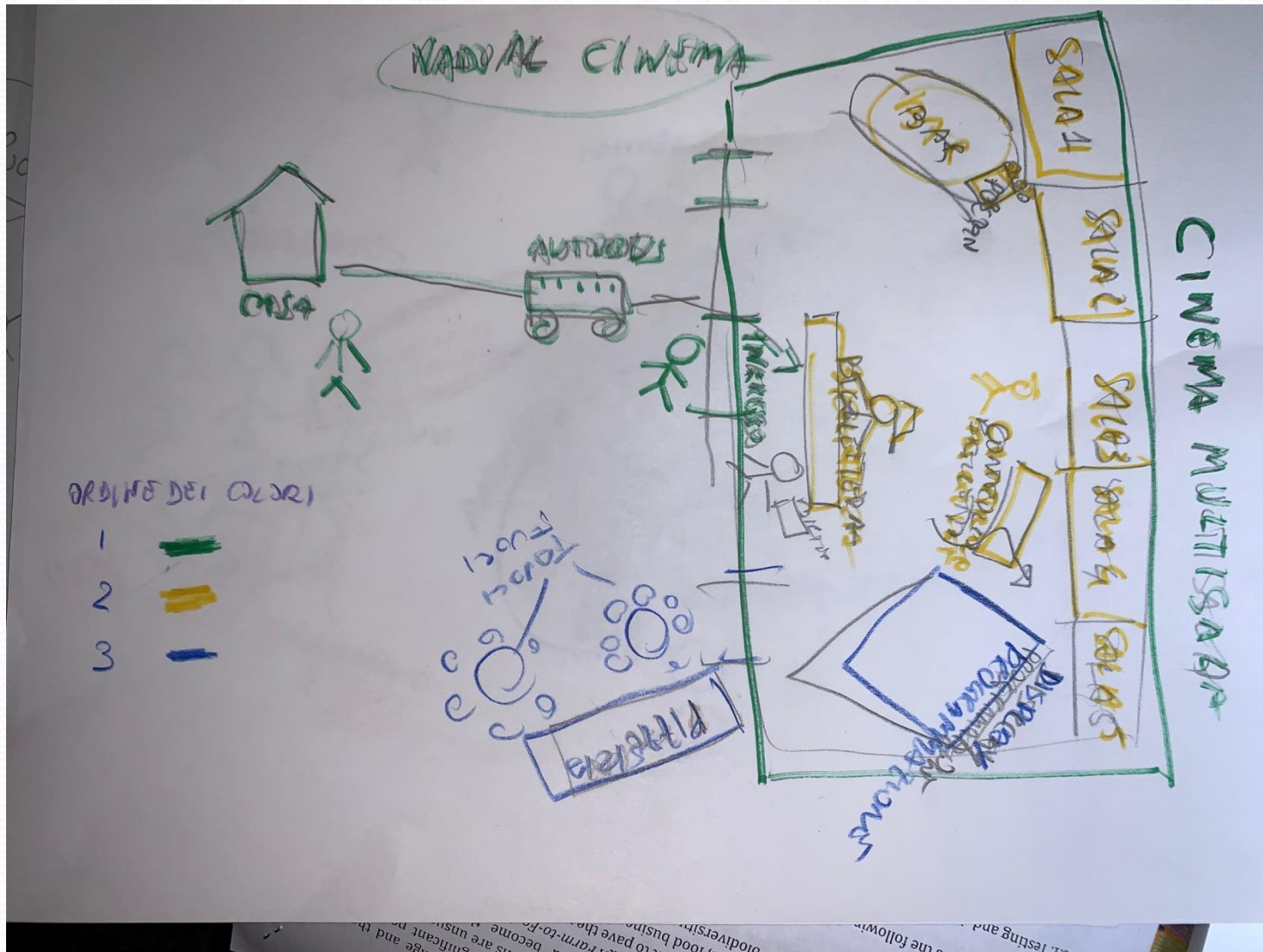
Prima parte

- Procuratevi 3 penne/matite/pennarelli di colori diversi
- Disegnate una mappa cognitiva per descrivere la vostra vita da studente universitario all'interno del campus di Fisciano
- Ricordate che una Cognitive Map può riferirsi a un edificio o a uno spazio – ma può anche essere un processo, come quello di svolgere un compito assegnato dal docente o prendere un libro in prestito dalla biblioteca.
- Avete circa 6 minuti per disegnare e ogni 2 minuti vi chiederò di cambiare colore.
- Scrivete che ordine di colori avete usato e completate le etichette associate alle caratteristiche disegnate
- Caricate il file con la foto del disegno su el-platform

Lo scenario di Mike Ross disegnato come cognitive map (Esempio)



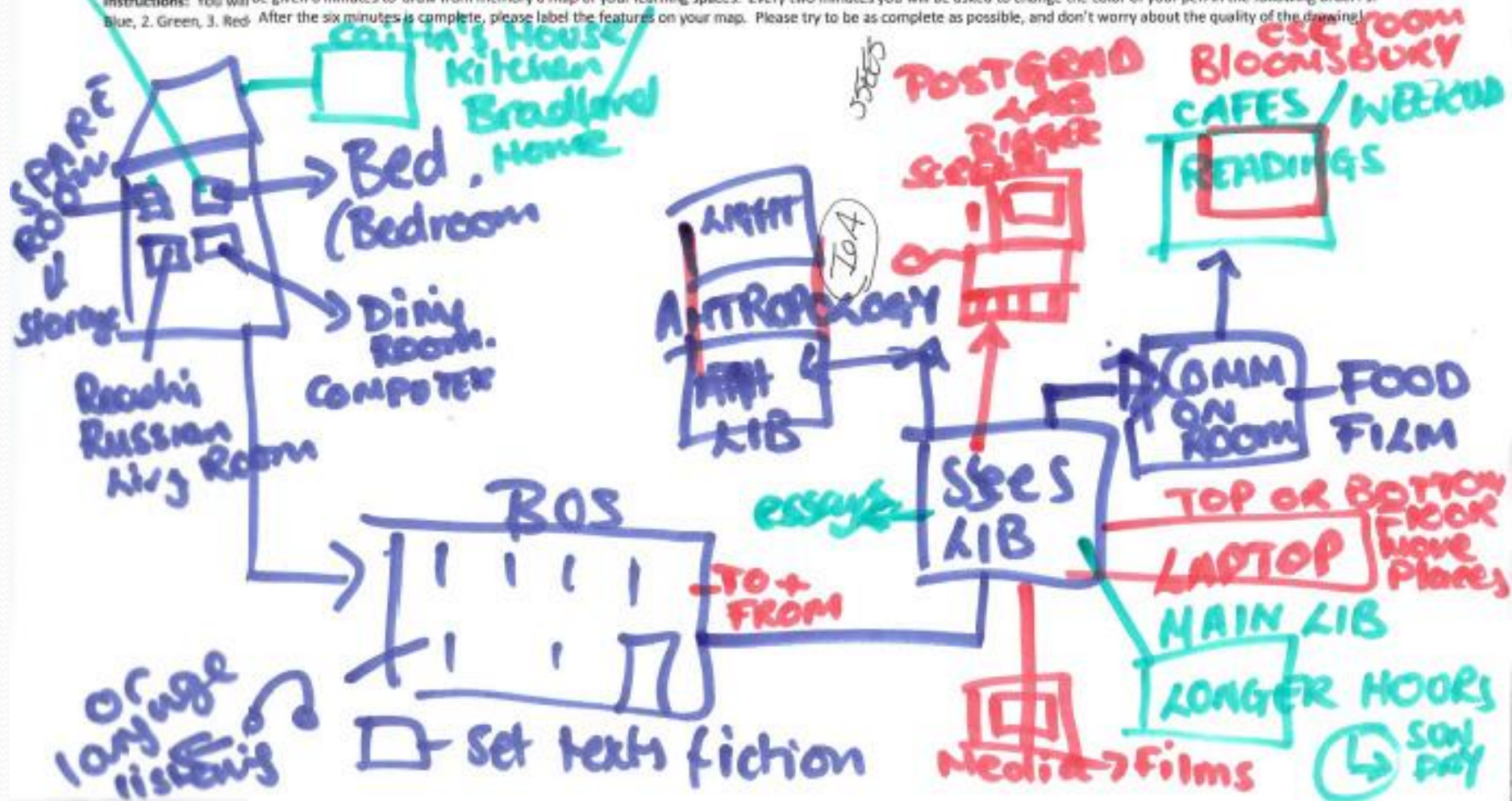
La cognitive map «vado al cinema»(Esempio)



ESSAYS - LONGER SILENT 24h CONCENTRATION

Instructions: You will be given 6 minutes to draw from memory a map of your learning spaces. Every two minutes you will be asked to change the color of your pen in the following order: 1. Blue, 2. Green, 3. Red. After the six minutes is complete, please label the features on your map. Please try to be as complete as possible, and don't worry about the quality of the drawing!

UCL 22 f SSes MA



Esercizio sulle cognitive maps

Seconda parte

- Ciascuno di voi analizza la mappa cognitiva di un collega e scrive una breve considerazione su ciò che sembra essere importante dal punto di vista della persona coinvolta nell'indagine di contesto (lo studente magistrale di UniSa)
- Caricate la foto e il commento alla foto